

Shree Pandit

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EDUCATION

Maulana Azad National Institute of Technology, Bhopal	Expected May 2028
B.Tech - Computer Science and Engineering	CGPA: 7.40
The Himalaya Academy, Dewas	2023
Madhya Pradesh Board of Secondary Education – Class XII	Percentage: 85.2%
Carmel High School, Gadchiroli	2021
Central Board of Secondary Education – Class X	Percentage: 93.8%

WORK EXPERIENCE

Data Engineering Intern	May 2026 – July 2026
Transform Rural India Foundation (TRIF)	
- Centralized 186,000+ unstructured survey responses into 7 structured fact tables by architecting a grain-aware star-schema ETL pipeline in Python (Pandas, NumPy) that rectified schema inconsistencies and encoding errors. - Accelerated KPI reporting from weekly manual exports to near-real-time for 2,000+ Gram Panchayats across 5 states by designing a multi-page Power BI dashboard tracking 150+ metrics. - Cut manual report generation time by 80% for BLCC Fact Sheets by automating end-to-end web scraping workflows via Python (Selenium).	

PROJECTS

A.R.G.U.S C++, Python, FastAPI, React, Tailwind, ZeroMQ, TimescaleDB, Redis	GitHub
AI-driven Recognition Gateway for Unidirectional IP Security in Critical Infrastructure Networks.	
- Enabled passive threat detection on air-gapped, unidirectional networks by engineering a C++ DPI engine to extract flow-level security features (JA3 fingerprints, DNS entropy, host fan-out profiling) with zero return-path capability. - Bridged real-time ML inference into a live SOC dashboard by architecting a ZeroMQ PUB/SUB pipeline and a schema-translating consumer service, resolving alert-payload mismatches to ensure zero data loss between the detection and visualization layers. - Delivered alert broadcasting with automatic Redis-failover recovery by building a FastAPI backend with Redis Pub/Sub and WebSocket fan-out, backed by a fail-closed PostgreSQL/TimescaleDB (SQLite-fallback) persistence layer.	

DPI Engine C++20, Python, FastAPI, React.js, Docker	GitHub
Multi-threaded Deep Packet Inspection engine with a full-stack web interface for PCAP traffic analysis.	
- Classified 22+ applications from raw traffic in real time by engineering a multi-threaded C++20 DPI core with FiveTuple flow tracking and TLS SNI extraction. - Blocked malicious flows at the connection level by implementing rule-based IP/domain evaluation with an HTTP Host fallback mechanism to drop unauthorized packets. - Processed 50 MB .pcap files in under 10 seconds by architecting a Dockerized FastAPI backend and React frontend to orchestrate analysis and visualize app distributions.	

Auralis C++, Python, FastAPI, React.js, Npcap, WebSockets	GitHub
Real-time network traffic diagnostic instrument featuring cross-platform C++/Python IPC.	
- Achieved sub-10ms parsing of raw L2/L4 network frames by engineering a C++ capture engine (libpcap/Npcap) streaming 5-tuple flow data via a custom TCP IPC bridge. - Delivered zero-latency metrics with no database persistence by architecting an async Python (FastAPI) aggregation layer using a 300-second in-memory ring buffer for flow tracking. - Rendered live bandwidth and connection states at 1-second intervals by developing a React/Tailwind dashboard with an SVG oscilloscope trace and a WebSocket-driven patch-panel visualization.	

NITBFreshers Study Portal JavaScript, Tailwind CSS, PHP, MySQL	GitHub
Full-stack academic resource management system for MANIT students.	
- Supported 1,000+ concurrent users (95% batch adoption) by deploying a multi-instance PHP/MySQL backend. - Eliminated SQL injection risk across the platform by enforcing prepared statements and 64-character HttpOnly authentication tokens. - Cut load times by 40% and secured 100% of academic resources against unauthorized downloads by replacing synchronous reloads with async Fetch API calls and custom PDF.js-based rendering.	

CODING PROFILES

LeetCode: ShreePandit (2170, Guardian)	CodeChef: shree_pandit (1815, 4-Star)	Codeforces: ShreePandit (1610, Expert)
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TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, PHP
Web & Frameworks: React.js, Next.js, FastAPI, Node.js, Express.js, Tailwind CSS
Data Libraries: NumPy, Pandas, Selenium, Matplotlib, Seaborn
Databases: PostgreSQL, MySQL, MongoDB, TimescaleDB, Redis
Tools & Platforms: Git, Docker, ZeroMQ, Microsoft Power BI
CS Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Artificial Intelligence, Data Science

ACHIEVEMENTS

- 2nd Runner-Up at Kinetic Coding Challenge by IIPE Visakhapatnam.
- National Finalist at Shaastra Programming Contest by IIT Madras.
- National Finalist at Code Wars by IIT Jodhpur at Prometeo' 26.
- Secured Global Rank 23 at Code Quest 2.0 by IIIT Kottayam.

POSITION OF RESPONSIBILITY

Corporal, 12 MP BN NCC, MANIT Bhopal	August 2024 – Present
Team Management Head, ISTE SC MANIT	February 2025 – Present